SimpsonEF: EXPLAINABLE EJECTION FRACTION FROM PREDICTED SONOGRAPHER TRACINGS AND SIMPSON'S RULE

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ABSTRACT

Video networks estimate left-ventricular ejection fraction (EF) well but return a number whose derivation cannot be checked. SimpsonEF computes EF the way a sonographer does: it predicts the volume tracing (long axis and 20 chords) at end-diastole (ED) and end-systole (ES) of every beat and applies Simpson's method of disks, so the reported EF is a fixed formula of drawn, inspectable geometry. Three small networks trace every frame (FrameTracer), find the beats on the resulting volume curve (BeatFinder) and jointly redraw each beat's ED and ES tracings (BeatTracer), the last trained through Simpson's formula in EF units. On all 1277 EchoNet-Dynamic test videos, with no annotation at inference, SimpsonEF reaches R^2 0.804 and MAE 4.02 EF points with 4.5M parameters: non-inferior to the video regressors we re-ran on the same videos, and 0.16 above previously reported keypoint-derived EF. Because the output is a formula, it can adopt a new dataset's measurement convention: zero-shot on CAMUS, whose label is a biplane Simpson EF, it reaches R^2 0.710 (95% CI 0.49–0.84), where eight released EF models reach at most 0.28. Conformal intervals cover 90.3% of test cases, and the tracings segment the ventricle with Dice 0.935/0.915.

Index Terms— Ejection fraction, echocardiography, Simpson's method, keypoints, explainability, conformal prediction

1. INTRODUCTION

Left-ventricular ejection fraction (LVEF, here EF), the fraction of the end-diastolic volume ejected in a beat, is the principal measure of systolic function [1]. Video networks estimate EF to about 4 EF points [2, 3] but return a bare number. Clinicians compute EF from a tracing by Simpson's method of disks, biplane where possible [4], so a model whose output is that tracing, and whose EF is that formula, can be checked frame by frame.

Related work. Direct regression with 3D CNNs [2], CNN–transformers [5], contrastive regression [3] or light spatio-temporal networks [6] reaches R^2 0.81–0.82. Foundation models pretrained on millions of echocardiograms [7, 8, 9, 10] give EF estimates but are black boxes with tens of millions to 1.1B parameters. Explainable models use graph attention [11] or prototypes [12] but still regress EF; uncertainty-aware ones use Bayesian networks [13] or conformal prediction [14]. Segmentation pipelines derive EF from predicted masks [15], although explicit masks do not improve learned EF regression [16]. Keypoint models come closest to the clinical computation: EchoTrace [17] regresses the 84 tracing coordinates of single frames, and EchoGraphs [18] finds contour keypoints with a graph network but reports that Simpson's EF from them (R^2 0.65 on whole videos) trails a black-box regressor (0.79).

We contribute: (i) a video-to-EF pipeline whose output is a fixed Simpson computation on predicted tracings, with every intermediate inspectable; (ii) joint ED/ES tracing trained through the formula in EF units, which closes the MAE gap to direct regression without degrading the anatomy; (iii) a like-for-like evaluation against eight released models on EchoNet-Dynamic and zero-shot on CAMUS, covering EF, segmentation, ED/ES detection and uncertainty; and (iv) an analysis of where formula EF loses accuracy.

2. METHOD

Tracings and EF. A tracing has the ends A_1, A_2 of the long axis and the ends L_i, R_i of 20 perpendicular chords (EchoNet-Dynamic convention). With $d_i = \|L_i - R_i\|$ and $\ell = \|A_1 - A_2\|$,

$$V = \frac{\pi}{4}\,\frac{\ell}{20}\sum_{i=1}^{20} d_i^2, \qquad \mathrm{EF} = 100\Big(1 - \frac{V_{\mathrm{ES}}}{V_{\mathrm{ED}}}\Big). \tag{1}$$

No mask is trained on or enters the EF; masks only measure Dice.

Tracers. FrameTracer and BeatTracer are fully convolutional encoder–decoders (instance norm, GELU, skip connections, dilated bottleneck, no global pooling) with one full-resolution heatmap per point; each point is the soft-argmax [19] over a 7×7 window at the peak. The same weights run at scales 1, 1/2 and 1/4 and the logits are fused by log-sum-exp, so FrameTracer's VAL Dice changes by at most 0.011 between 0.5× and 3× the training size. FrameTracer (1.42M parameters) takes 13 frames (offsets −24..24, step 4) and returns 42 points. BeatTracer (2.53M, width 32) takes the same 13-frame windows around ED and around ES stacked as channels and returns both tracings (84 points), so ES is drawn knowing the ED appearance.

Beats. BeatFinder, a two-layer bidirectional GRU (0.53M), reads FrameTracer's chord sequence $x_t = (d_1, \ldots, d_{20}, \ell)$ and outputs ED and ES heatmaps over time (squared error to Gaussians around the annotated beat) and EF quantiles (pinball loss; used for the interval and to pick BeatFinder's epoch, never as the EF). Peaks give each beat's ED and following ES (if none pair, the extremes of FrameTracer's volume curve). It only chooses frames, checkable on the volume curve.

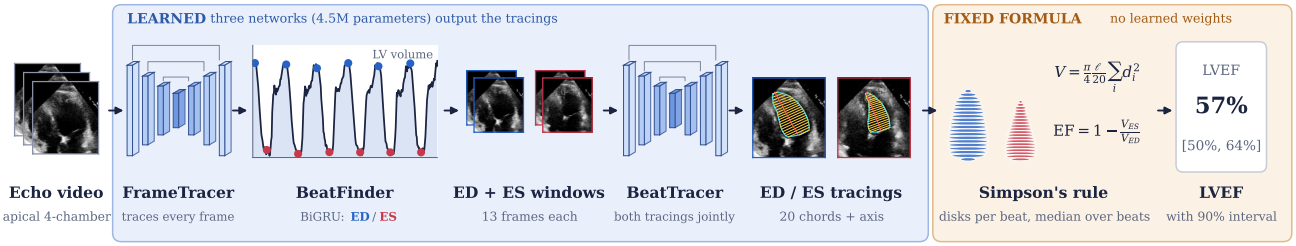


Fig. 1. SimpsonEF on a TEST case. Learned (blue): FrameTracer traces every frame, BeatFinder marks each beat's ED and ES on the volume curve, BeatTracer redraws both tracings from the ED and ES windows. Fixed (orange): Simpson's rule turns the chords into volumes; LVEF is the median over beats, with a 90% interval.

Training through the formula. Both tracers use an L_1 point loss, heatmap cross-entropy (σ=1.5 px) and L_1 losses on $\log(V+1)$, on its ED→ES change and on the log chord widths weighted by their share of V (weights 1.5, 1, 1.06, 4, 2; ES frames ×1.25), with similarity transforms (rotation ±20°, scale 0.8–1.2, shift ±18 px, frame rescaled 0.5–2.5× on its canvas), edge occlusion and scanner-style degradations (gain, gamma, speckle, noise, blur, shadows). Its ED→ES change is $\approx \log(1-\mathrm{EF})$, so an EF point costs $\approx 1/(1-\mathrm{EF})$: 1.4 at EF 30% but 3.3 at 70%, i.e. this loss values low-EF hearts least. BeatTracer is therefore fine-tuned (12 epochs, learning rate 10^{-4}, no dropout) adding $40\mathscr{L}_{\mathrm{EF}}$, $\mathscr{L}_{\mathrm{EF}} = w_b\,|\mathrm{EF}(\hat{P}_{\mathrm{ED}}, \hat{P}_{\mathrm{ES}}) - \mathrm{EF}^*|$ in EF units (fractions), $w_b \propto$ the inverse square root of the TRAIN frequency of the label's EF bin (<20, 20–29, …, ≥80%; mean 1). BeatTracer is trained without the scanner-style degradations, since input noise shrinks outputs towards the mean (attenuation bias [20]).

Inference and uncertainty. BeatTracer is applied to 15 rescaled (0.85–1.17) and rotated (±8°) copies of each beat's input and the mapped-back points are averaged; (1) gives the beat's EF and the median over beats is reported. A normalised split-conformal interval [21, 22, 23] uses the scale $s = (\hat{q}_{.95} - \hat{q}_{.05})/2 + |\widehat{\mathrm{EF}} - \widehat{\mathrm{EF}}_{\mathrm{GRU}}|$, where BeatFinder also predicts EF quantiles from FrameTracer's chords: s grows when the clip is uncertain or the two routes to EF disagree. With Q the conformal quantile of $|y - \widehat{\mathrm{EF}}|/s$ on VAL (n=1288), $\widehat{\mathrm{EF}} \pm Qs$ covers 90% of exchangeable cases.

Interpretability. The reported EF is computed only by (1); BeatFinder's learned EF estimate only widens the interval. A clinician can inspect and correct every intermediate: the chosen frames, tracings, chords, per-beat EF, and the cause of a wide interval.

3. EXPERIMENTS

Protocol. EchoNet-Dynamic [2] has 7465/1288/1277 TRAIN/VAL/TEST videos, each with an EF label and, except five TRAIN videos and one TEST video, one annotated beat (ED/ES frames and tracings). Training uses the 7460 annotated TRAIN videos (minus 81 records with several tracings stacked on one frame; frame windows are clamped at clip boundaries); EF, which needs only the video and its label, is scored on all 1277 TEST videos, Dice and ED/ES detection on the 1276 annotated ones. The final configuration was selected on VAL: the tracer variant by full-pipeline VAL R^2 among 25 trained variants (the top two tied within 0.001; we kept the simpler one), then input-copy averaging and aggregation over beats; TEST was run once on the frozen pipeline. VAL also calibrates the intervals, so it serves both selection and calibration. At test time no annotation is used. Metrics: R^2, MAE (EF points) and Pearson r against the EF label; Dice of the mask drawn from a tracing. The tracers are trained from scratch (AdamW, learning rate 3×10^{-4}, weight decay 10^{-4}, cosine decay, EMA 0.999, dropout 0.1, 32 frames per batch, 40 epochs, then BeatTracer's EF stage) and BeatFinder for 16 epochs (learning rate 10^{-3}); one script rebuilds all three networks with fixed seeds; rebuilding BeatTracer from scratch with another seed gives R^2 0.816 on VAL and 0.799 on TEST (0.820 and 0.804 here). Inference takes 0.49 s per 183-frame video on one RTX 3090 (0.36 s without input-copy averaging).

Comparison with released models. We ran every model whose weights are public (R in the tables) on the same 1277 EchoNet-Dynamic TEST videos and, unchanged, on the 50 CAMUS test patients; models without public weights are listed with the authors' figures (P). Each model uses its official preprocessing, with all clips of a video averaged; foundation models are used zero-shot or as a frozen encoder with a head fitted on EchoNet TRAIN (ridge for EchoFM, its attentive probe for EchoJEPA).

EF. Formula EF reaches R^2 0.804 (95% CI 0.778–0.828), MAE 4.02 and r 0.897 (Table 1), 0.16 above the keypoint-derived EF reported by EchoGraphs on whole videos. Against the EchoNet-trained regressors on the same videos, the paired bootstrap MAE difference is −0.01 (−0.18, +0.16) for EchoNet-Dynamic and +0.09 (−0.08, +0.25) for Echo-E^{3}Net: SimpsonEF is non-inferior at a 0.3-point margin, not better; it beats EchoGNN (−0.45) and the foundation models (0.63 at best). Reduced EF is detected with AUROC 0.978 at EF <40% (EchoNet-Dynamic and Echo-E^{3}Net 0.982; sensitivity/specificity 71.9/98.4% vs 70.6/98.1% and 78.8/98.2%) and 0.956 at <50%. MAE 4.0 is smaller than the 7.6–13.9-point inter-observer variation cited in [2].

Segmentation. Against EchoNet's own masks, BeatTracer's tracings reach Dice 0.935/0.915 (ED/ES) against 0.920/0.897 for the released DeepLabV3 [24, 2] (Table 2, Fig. 2); against the released PNG masks, which include the long-axis ends, 0.917/0.907 against 0.903/0.887. Published mask-trained segmenters reach 0.936–0.940/0.916–0.923 with 7–56M parameters (BeatTracer: 2.5M); given other evaluation code and our output sharing the chord representation of the reference masks, we claim parity, not superiority.

ED/ES detection. BeatFinder, which unlike dedicated detectors [25, 26] learns from the one annotated beat, places ED and ES within 1.78 frames on average, 76.4% of them

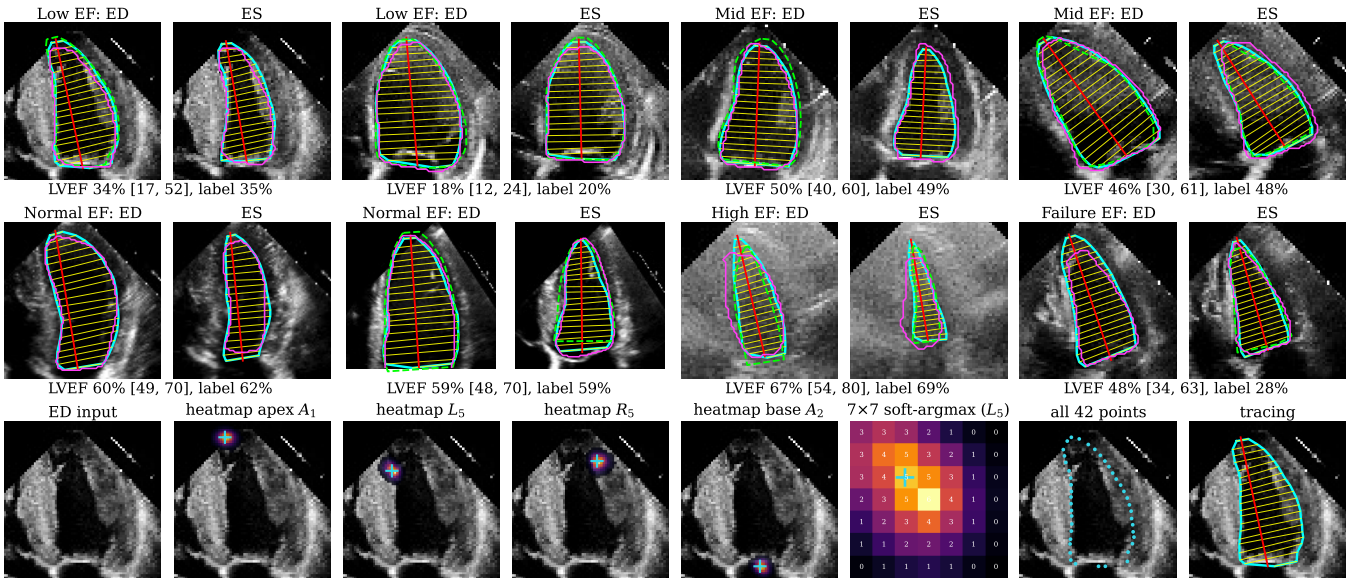


Fig. 2. Top rows: TEST cases (ED left, ES right): BeatTracer's tracing (cyan; chords yellow, axis red), DeepLabV3 mask (magenta), reference (green dashed); formula EF, 90% interval and label (last: a failure). Bottom row: how BeatTracer turns heatmaps into the tracing (first case, ED): one heatmap per point (4 of 42 shown), a soft-argmax over the 7×7 window at the peak (weights in %) gives the sub-pixel point (+), and the 42 points form the tracing.

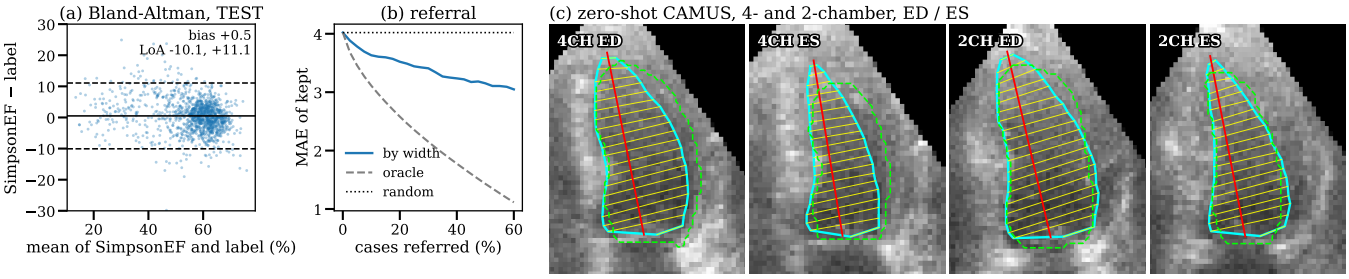


Fig. 3. (a) TEST: Bland–Altman of SimpsonEF against the label. (b) MAE after excluding the widest intervals. (c) Zero-shot CAMUS: FrameTracer's tracing (cyan) and the CAMUS reference (green dashed).

within 2 frames (Table 4); the extremes of FrameTracer's volume curve alone give 2.85 (57.1%), so BeatFinder adds value; EchoGNN's attention gives 3.43 on the videos where it is determinable. On CAMUS half-sequences the error is 2.63 frames (47% within 2).

Training objective. Table 5 shows the steps on VAL (differences below ≈0.01 are within seed variation); the EF-unit formula loss is the largest gain, and Simpson's formula on BeatTracer's tracings outperforms BeatFinder's learned EF on the same chords (0.810 vs 0.820). The loss does not buy EF with worse anatomy: on VAL, Dice (0.916/0.906 before, 0.914/0.905 after) and absolute volume errors (ED 13.2% to 12.7%, ES 16.2% to 15.4%) hold or improve. ED and ES volume errors are correlated (0.79 before, 0.80 after): a shared scale error, present without the EF loss, that cancels in the ratio.

Where the error is. Simpson's rule on the sonographer's own TEST tracings reproduces the label (R^2 0.991 on the 1264 videos with one tracing per frame; 0.943 with the 12 whose tracings are stacked), so the error is in the predicted tracing, and there in the chord widths: at the annotated frames BeatTracer's tracings give 0.790, with the reference widths 0.969, with the reference axis only 0.805. The residual is EF compressed towards normal (slope 0.81; Bland–Altman [34] bias +0.5, limits −10.1 to +11.1, Fig. 3a): +3.4 to +5.5 points below EF 50%, +0.8 to −1.4 in 50–70%, −5.7 above 70%. Tail oversampling flattens it (slope 0.87); no variant tried, including width-targeted losses and longer pretraining, exceeds VAL R^2 0.821. At the annotated frames, replacing both tracings by the reference recovers the label (MAE 0.84), but replacing only the worse one does not help (4.03 to 4.69), because the shared error then no longer cancels. Of the 87 TEST errors above 10 points, 55 persist at the annotated frames, 10 stem from beat selection; in the failure of Fig. 2 the ED tracing is 19% too large and the ES 18% too small, errors that do not cancel.

Uncertainty. Calibrated on VAL, the intervals cover 90.3% of TEST cases (95% CI 88.3–91.6) with mean width 16.7 (±8 points, wide clinically), and 87.5% of cases below EF 40% (constant width: 73.1%). Width tracks error: it detects errors above 10 points with AUROC 0.73, and excluding the 20% widest intervals lowers the MAE of the rest from 4.02 to 3.52 (Fig. 3b).

Zero-shot CAMUS. CAMUS [35] differs in protocol, not

Model	Src	Par. (M)	Training data	MAE	R^2
SimpsonEF (ours)	R	4.48	EchoNet	4.02	0.804
EchoNet-Dynamic [2]	R	31.3	EchoNet	4.01	0.815
Echo-E^{3}Net [6]	R	1.76	EchoNet	3.92	0.819
EchoGNN [11]	R	1.7	EchoNet	4.46	0.758
EchoCoTr-B [5]	P	21	EchoNet	3.98	0.81
CoReEcho [3]	P	21	EchoNet	3.90	0.82
EchoGraphs [18]	P	27.6	EchoNet	4.23	0.79
EchoGraphs, Simpson [18]	P	27.6	EchoNet	4.30	0.65
PanEcho [8]	R	42.0	Yale (0-shot)	5.47	0.600
EchoPrime [7]	R	123.1[a]	Cedars (0-shot)	7.78	0.210
EchoCLIP [27]	R	151.5	Cedars (0-shot)	11.51	−0.387
EchoFM-L [28] (probe)	R	344.5[b]	290K clips, probe	5.64	0.627
EchoJEPA-L [10] (probe)	R	303.9	18M echos, probe	5.48	0.630

Table 1. EF on all 1277 EchoNet-Dynamic TEST videos. R: released weights run by us; P: published figures. Par.: parameters (M). Foundation models are zero-shot (0-shot) or a frozen encoder with a head fitted on EchoNet TRAIN only (probe). SimpsonEF: R^2 95% CI 0.778–0.828, MAE 3.82–4.22. [a]Encoder and view classifier. [b]Full masked-autoencoder checkpoint.

Model	Src	Par.	ED	ES	Pool
SimpsonEF (ours)	R	2.53	93.5	91.5	92.5
DeepLabV3 [2]	R	39.6	92.0	89.7	90.9
nnU-Net [29]	P	7.37	93.6	91.6	92.9
SepXception [30]	P	55.8	93.6	91.7	92.9
UniLVSeg [31]	P	18.8	94.0	92.3	93.3
MSV-Mamba [32]	P	n.r.	93.7	91.7	92.9

Table 2. LV segmentation on EchoNet-Dynamic TEST: Dice (%) at the annotated ED and ES frames (1276 annotated videos). R: ours, against EchoNet's own masks (40 chord points); against the released PNG masks SimpsonEF scores 91.6/90.7 and DeepLabV3 90.3/88.7. P: authors' figures.

(a) EF

Model	MAE	R^2
SimpsonEF (ours)	5.08	0.710
EchoNet-Dynamic	9.71	0.079
Echo-E^{3}Net	11.21	−0.303
EchoGNN	9.94	−0.178
PanEcho	7.53	0.283
EchoPrime	18.63	−1.832
EchoCLIP	9.33	0.072
EchoFM-L (probe)	9.10	0.100
EchoJEPA-L (probe)	10.57	−0.099

(b) LV segmentation, Dice (%)

Model	ED	ES	Pool
SimpsonEF (ours)	90.5	86.8	88.7
DeepLabV3	88.1	79.9	84.0

Table 3. Zero-shot CAMUS test (50 patients). (a) EF against the biplane label (ours: biplane Simpson; others: mean of their 4- and 2-chamber EF). (b) 4-chamber LV Dice (%).

Model	Src	Par.	ED	ES	Pool	Acc
SimpsonEF (BeatFinder)	R	1.95	1.74	1.82	1.78	76.4
EchoGNN [11][c]	R	1.7	2.75	4.15	3.43	44.0
Volume-curve extremes[e]	R	1.42	2.99	2.70	2.85	57.1
Latent motion [33][d]	P	n.r.	2.9	2.0	2.4	n.r.

Table 4. ED/ES detection on the 1276 annotated TEST videos: frame error and % within 2 frames (Acc). [c]Videos where determinable (87/83%). [d]Unsupervised. [e]FrameTracer's volume-curve extremes nearest the annotation.

only images: (i) its EF label is biplane Simpson (4- and 2-chamber views) with one disk height shared by ED and ES, whereas EchoNet labels are single-plane with per-phase heights; (ii) each view is a half cycle ED→ES (≈20 frames), not a multi-beat video; (iii) contours follow another convention. We run FrameTracer and BeatFinder unchanged on the 50 test patients, with no CAMUS data in training and no annotated frame, and apply the CAMUS label definition to the chords (its reference code reproduces the label, R^2 0.998). Which tracer to use was decided on the 400 CAMUS training patients, whose labels serve only this choice: biplane R^2 0.55 for FrameTracer, 0.51 for BeatTracer.

On the 50 test patients biplane formula EF reaches R^2 0.710 (95% CI 0.49–0.84; MAE 5.08, r 0.849; 4-chamber alone 0.567, 2-chamber alone 0.592; Dice 0.905/0.868 4CH, 0.893/0.864 2CH); with EchoNet's per-phase disk heights the same chords give $R^2 \approx 0$ (bias +9), so the label convention matters as much as the model. The rest is contour convention (Fig. 3c): our contours are narrower towards the base, most at ES; replacing only our 4-chamber tracings by the reference gives 0.79. A zero-shot model cannot learn this convention.

CAMUS against released models. The same models, run unchanged on CAMUS (Table 3), reach R^2 0.28 at best (PanEcho) and the EchoNet-trained regressors 0.08 or less, against 0.710 for SimpsonEF. Like for like on the 4-chamber view alone, SimpsonEF reaches 0.57 (0.18–0.78) against at most 0.22 (−0.26–0.49). The comparison favours a formula: the CAMUS label is itself a biplane Simpson EF, which a regressor cannot know, and CAMUS gives half a cycle per view (baselines see a synthetic cycle).

Table 5. Ablations on VAL (1288 videos, full pipeline, beats from BeatFinder); the final configuration was selected on these numbers. TEST, run once: R^2 0.804. *25-frame family (its reference: 0.809).

Variant	R^2	MAE	r	slope
FrameTracer tracings only	0.776	4.26	0.883	0.81
+ joint ED/ES tracing (BeatTracer)	0.797	4.06	0.894	0.78
+ EF-unit formula loss	0.813	3.95	0.904	0.82
+ width 32	0.820	3.89	0.906	0.83
+ input-copy averaging (SimpsonEF)	0.820	3.86	0.906	0.83
One change to SimpsonEF				
BeatFinder's learned EF instead of Simpson	0.810	3.87	0.900	0.81
mean instead of median over beats	0.791	4.08	0.890	0.82
no EF-unit loss	0.810	4.00	0.900	0.79
25-frame windows, 3 EF stages	0.821	3.89	0.907	0.82
tail-oversampled EF stage*	0.801	4.13	0.899	0.87

4. DISCUSSION AND CONCLUSION

Simpson's rule on predicted tracings is non-inferior to video regressors (paired MAE within ±0.3) with 4.5M parameters, every step inspectable, and adopts a new dataset's convention. Limitations: EF compressed at the extremes; few CAMUS patients; coverage under shift untested; no reader study. Code: github.com/gowthamaanp/simpsonef.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used open-access human subject data from EchoNet-Dynamic (Stanford University) and CAMUS (University Hospital of St Etienne); ethical approval was not required as confirmed by the licences attached to the data.

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